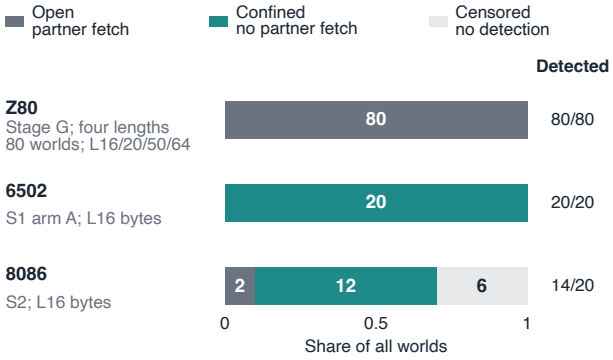
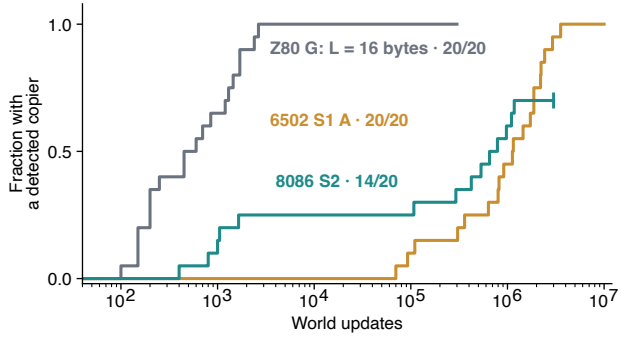
S1/S2: open ≥8/16 partner-fetch encounters; confined 0/16; no first cases in 1–7/16

**a** **First-replicator route**




**b** **Arrival: first detected heritable copier**




Observation ends: Z80 300,000; 6502 10,000,000;
8086 3,000,000 updates (6/20 undetected; censor tick).

**c** **Literal-channel expectation and observed beginnings**

| Machine | Literal width (bytes / push) | Exact-pusher candidate present (opcode motif) | First route among detected worlds |
|---|---|---|---|
| **Z80** | 2 | Yes · 01 c5 | Open · 80/80 |
| **6502** | 1 | No | Confined · 20/20 |
| **8086** | 2 | Yes · B9 51 | **12/14 confined; 2/14 open** |

Exact pusher: candidate writes its literal instruction bytes into an all-zero partner; routes: evolved worlds

**d** **8086 convergence: the 5F A4 x y first-copier family (10/14; all ten confined)**

| World seed | 5F A4 = POP DI; MOVSB | | | | | | | | x, y: variable bytes | | | | | | | | First detection (world updates) | Frozen full-address class (16/16 encounters) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | | |
| 8086001 | 5F | A4 | 63 | 3D | 5F | A4 | 63 | 3D | 5F | A4 | 63 | 3D | 5F | A4 | 63 | 3D | 652,500 | One-pass class |
| 8086005 | 5F | A4 | 73 | 5D | 5F | A4 | 73 | 5D | 5F | A4 | 73 | 5D | 5F | A4 | 73 | 5D | 975,000 | One-pass class |
| 8086006 | 5F | A4 | 6D | 7D | 5F | A4 | 6D | 7D | 5F | A4 | 6D | 7D | 5F | A4 | 6D | 7D | 1,650 | One-pass class |
| 8086009 | 5F | A4 | 61 | 3D | 5F | A4 | 61 | 3D | 5F | A4 | 61 | 3D | 5F | A4 | 61 | 3D | 292,500 | One-pass class |
| 8086012 | 5F | A4 | E2 | 7D | 5F | A4 | E2 | 7D | 5F | A4 | E2 | 7D | 5F | A4 | E2 | 7D | 1,100,000 | One-pass class |
| 8086013 | 5F | A4 | 7D | FD | 5F | A4 | 7D | FD | 5F | A4 | 7D | FD | 5F | A4 | 7D | FD | 1,175,000 | Loop class |
| 8086014 | 5F | A4 | E9 | 3C | 5F | A4 | E9 | 3C | 5F | A4 | E9 | 3C | 5F | A4 | E9 | 3C | 532,500 | One-pass class |
| 8086017 | 5F | A4 | 61 | BD | 5F | A4 | 61 | BD | 5F | A4 | 61 | BD | 5F | A4 | 61 | BD | 422,500 | One-pass class |
| 8086018 | 5F | A4 | 69 | DD | 5F | A4 | 69 | DD | 5F | A4 | 69 | DD | 5F | A4 | 69 | DD | 787,500 | One-pass class |
| 8086020 | 5F | A4 | 63 | FD | 5F | A4 | 63 | FD | 5F | A4 | 63 | FD | 5F | A4 | 63 | FD | 107,500 | Loop class |

Frozen class uses full instruction-start addresses: One-pass = no repeated start; Loop = repeated start.
Labels agree in all 16/16 encounters per tape. Post hoc: all ten repeatedly copy mapped ring positions at displacement d = 4 bytes.
G = historical random-code cohort; S1 A = 6502 primary arm; S2 = coherent-fetch 8086 arm.